

Auditing Historical Clinical Queries with Missing Availability Time

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What did the database know at 6 AM ?

HISTORICAL SNAPSHOT · 06:00

05:10	Heart rate	visible
05:40	Specimen order	visible
06:20	Lab result	not yet present
07:05	Clinical note	not yet present

What the model could query then



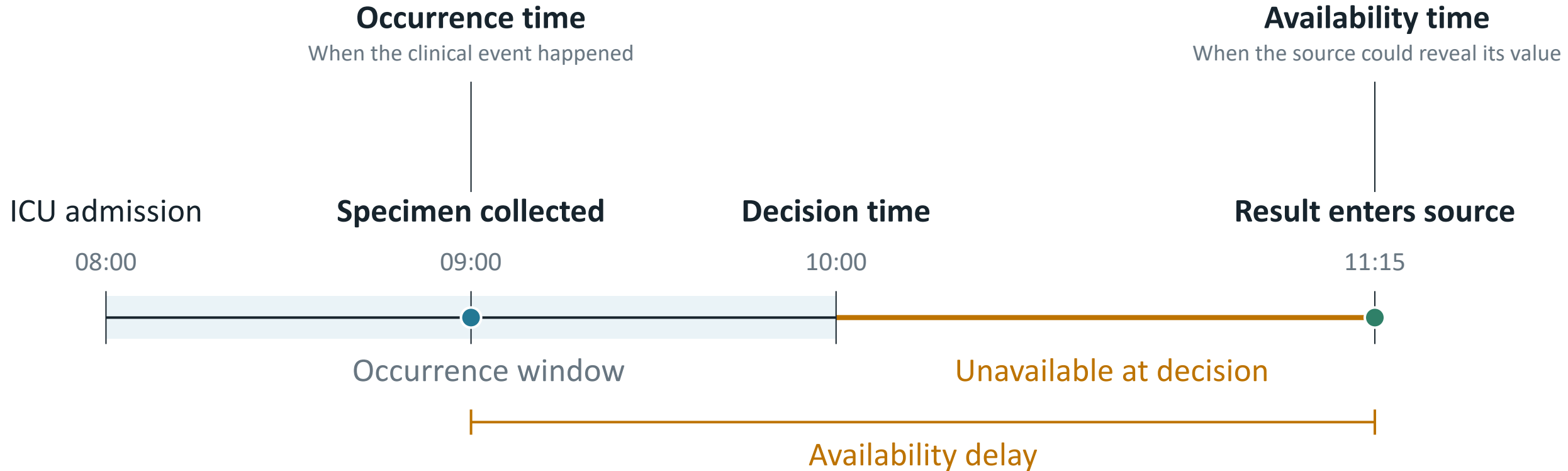
RETROSPECTIVE EXPORT · AFTER ENCOUNTER

05:10	Heart rate	present
05:40	Specimen order	present
06:20	Lab result	present
07:05	Clinical note	present

What became known eventually

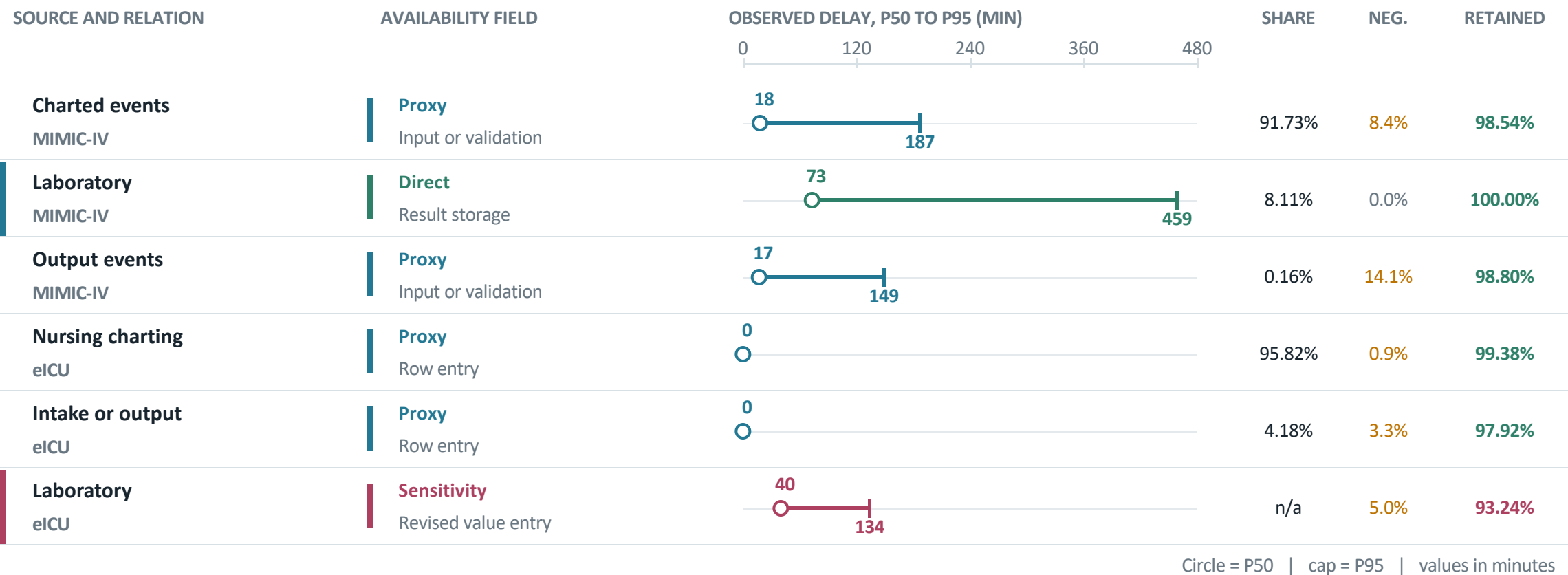
A historical query needs the database state then, not the completed record later.

One clinical event has more than one clock



Occurrence before the cutoff does not imply visibility before the cutoff.

Availability fields differ in meaning and delay



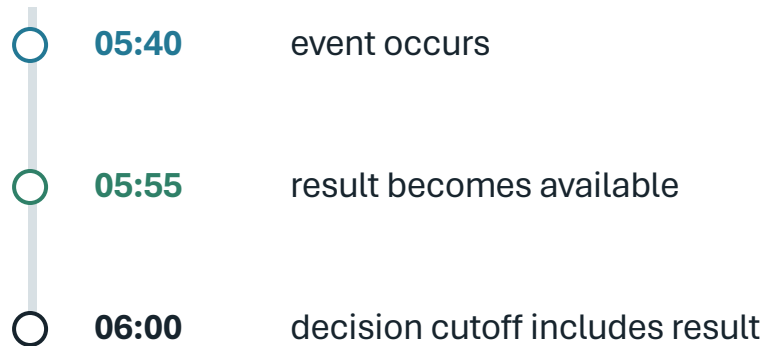
Missing availability leaves more than one history

RECORDED IN THE COMPLETED EXPORT

Occurrence 05:40 Result 7.1 mmol/L Availability unrecorded

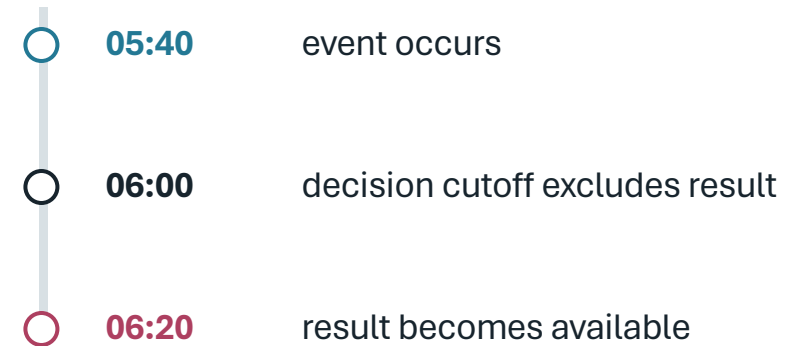
Compatible history A

Visible at 06:00



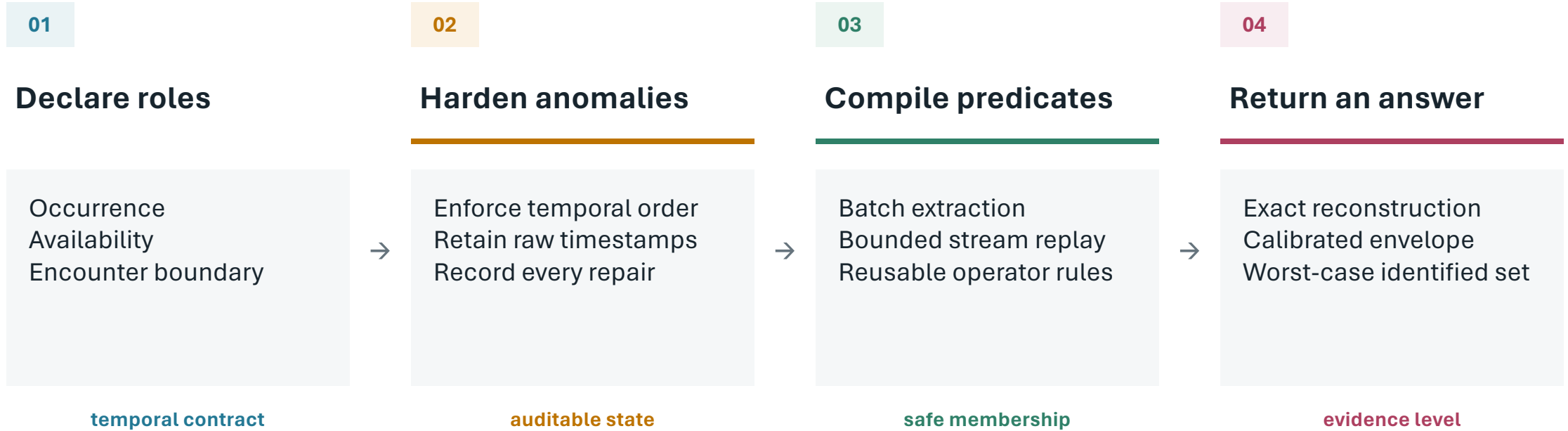
Compatible history B

Not visible at 06:00



The same completed record is consistent with different database states at 06:00.

Compiles contracts into historical queries



The evaluation uses two public ICU databases

Cohorts, source roles, and partitioning

Mapped numerical events from the first 24 h; landmarks at 1, 3, 6, 12, and 24 h

Design element	MIMIC-IV	eICU
Database version	3.1	2.0
Source cohort	94,458 ICU stays	200,859 unit stays
Primary relations	Charted; laboratory; output	Nursing; intake or output
Partitioning	Patient	Patient and hospital

SHARED SPLIT DESIGN

60·20·20

train / calibration / test

Five prespecified partitions

Research Questions

We evaluate historical reconstruction, uncertainty calibration and decision consequences

- ☐ *Does availability alter the reconstructed state?*
- ☐ *Does event coverage protect trajectories?*
- ☐ *What does simultaneous protection cost?*
- ☐ *When does calibration fail?*
- ☐ *Do point decisions hide ambiguity?*
- ☐ *Does the pattern survive broader scope?*

Does availability alter the reconstructed state?

Exact availability reconstruction at the 6 h cutoff

Occurrence-eligible events removed by the exact visibility predicate

Database	Relation	Excluded events (%)
eICU	Intake or output	11.9
MIMIC-IV	Output events	10.1
MIMIC-IV	Charted events	9.6
MIMIC-IV	Laboratory	8.6

FEATURE PROPAGATION

49.7%

of stays had at least one
changed feature

Maximum across evaluated horizons

Availability changes the reconstructed state before any model is fitted.

Does event coverage protect trajectories?

Marginal event coverage versus simultaneous all-source coverage

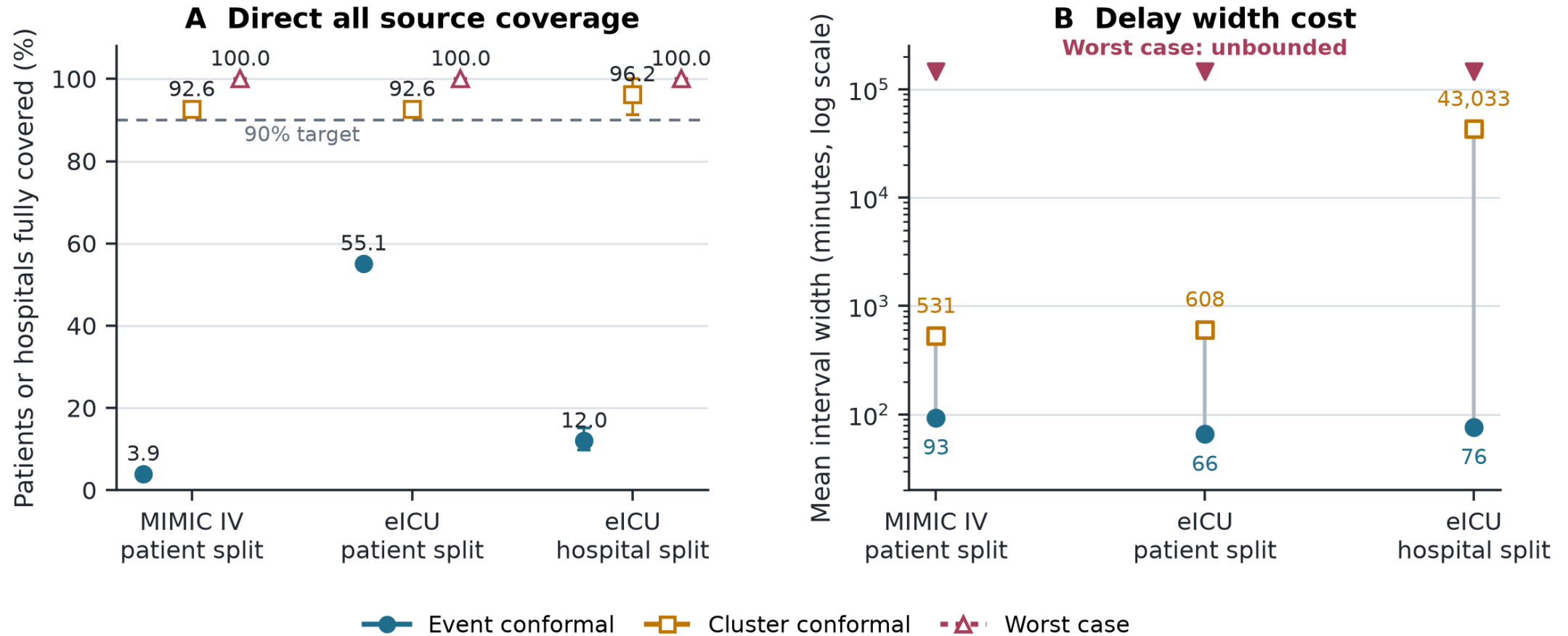
Means across five prespecified partitions; ranges are partition minima to maxima

Partition unit	Database	Mean target events per group	Event coverage (%)	All-source joint coverage mean (range), %
Patient	MIMIC-IV	224	90.2	3.9 (3.7–4.0)
Patient	eICU	143	90.2	55.1 (54.4–55.6)
Hospital	eICU	68,561	91.9	12.0 (9.8–15.2)

Marginal event coverage does not guarantee simultaneous trajectory coverage.

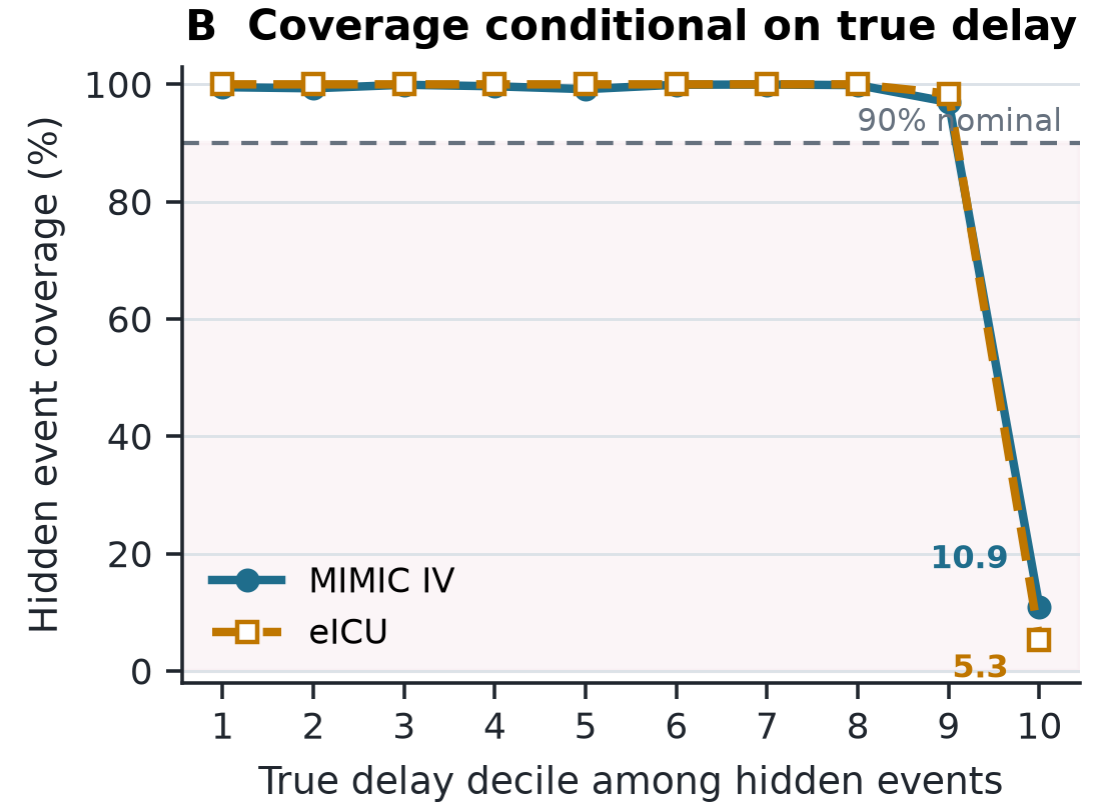
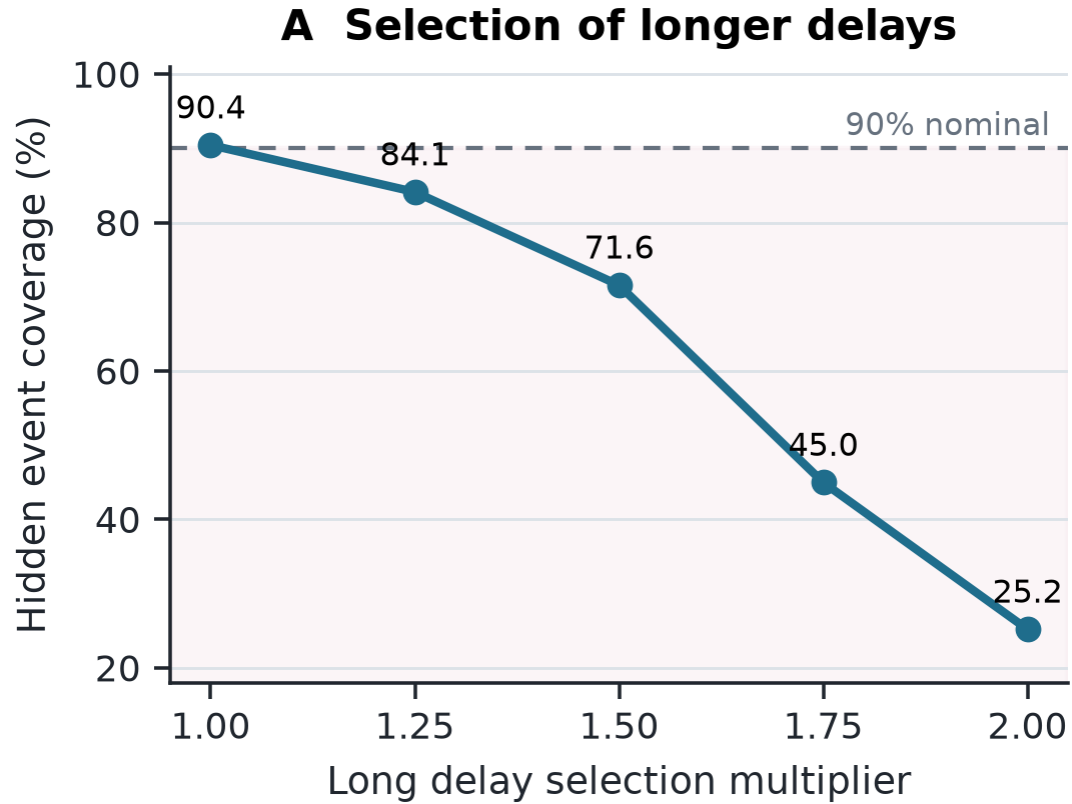
What does simultaneous protection cost?

Coverage and information across the identification ladder



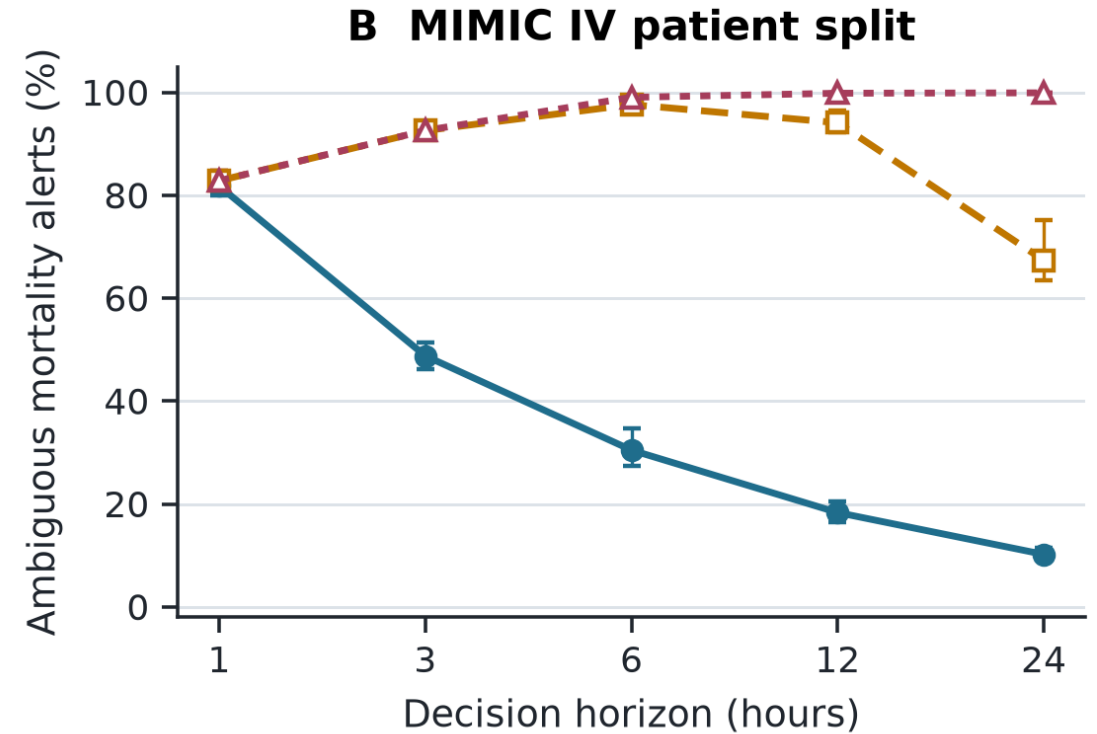
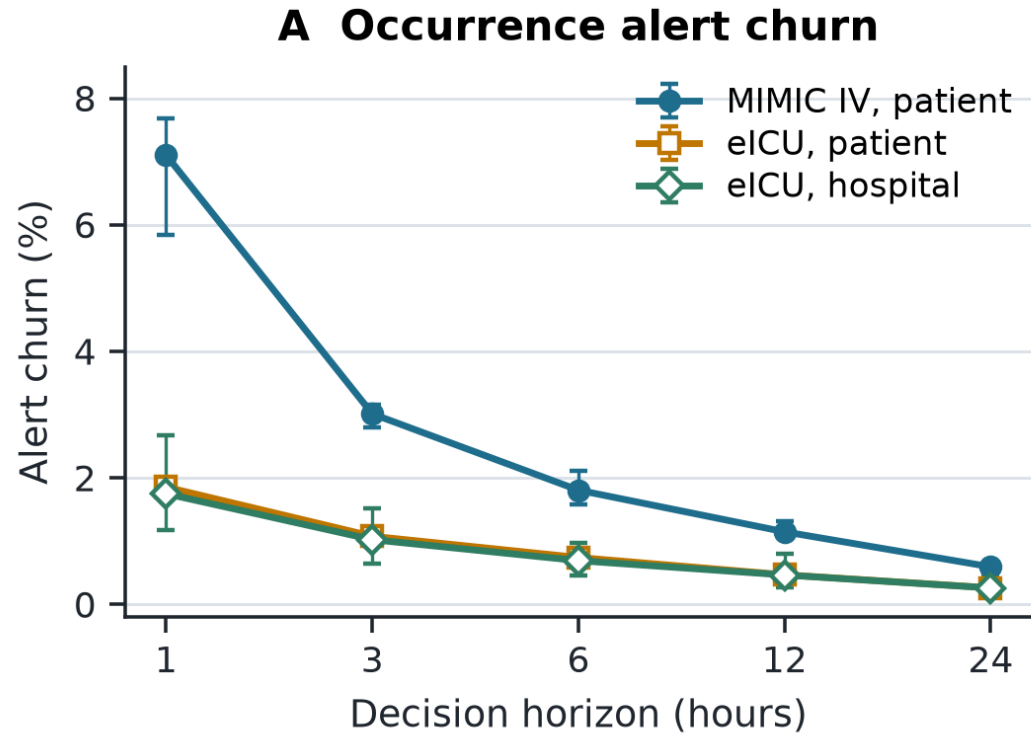
Simultaneous protection is achievable, but temporal precision can collapse.

When does calibration fail?



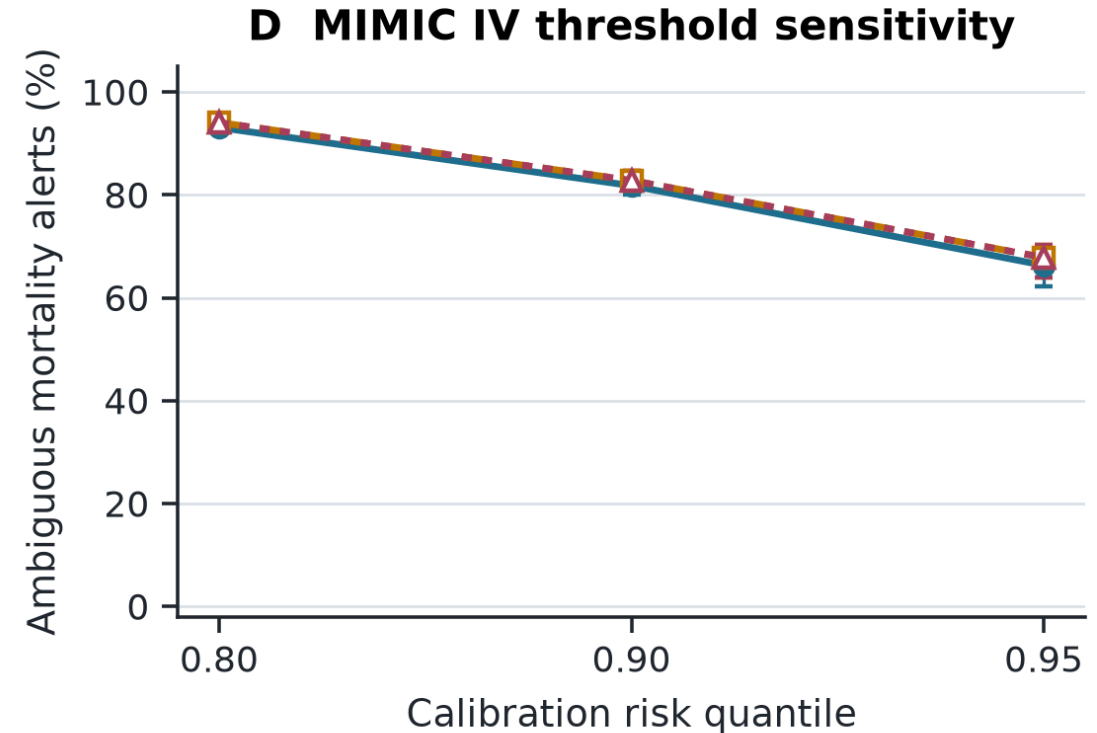
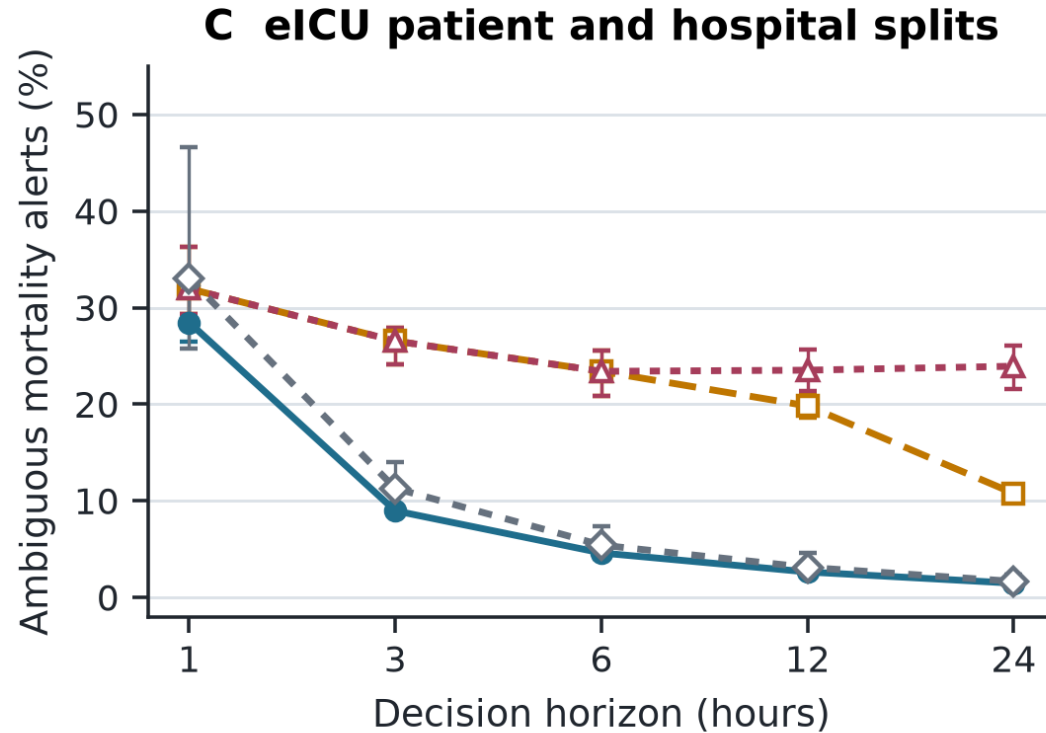
Calibration fails when missing delays are no longer comparable with calibration delays.

Do point decisions hide ambiguity?



Agreement between two points is not stability over the admissible set.

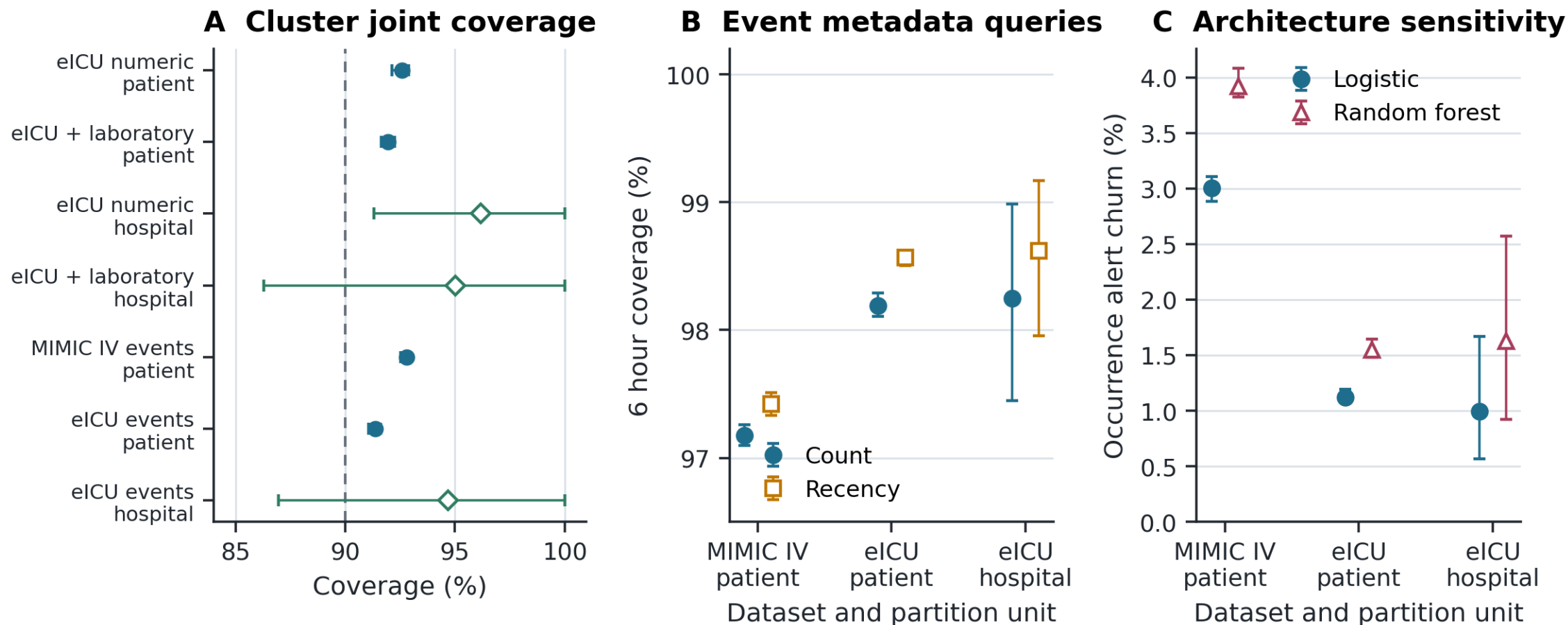
Do point decisions hide ambiguity?



—●— Event conformal —□— Cluster conformal -△- Worst case -◇- Event conformal, hospital

Agreement between two points is not stability over the admissible set.

Does the pattern survive broader scope?



The evidence ladder persists, while clinical utility remains outside the study's scope.

Takeaways

01

Source timestamps are not interchangeable.

Audit field meaning before measuring delay.

02

Marginal calibration is not trajectory protection.

Match calibration and query units.

03

Missing availability creates a set of histories.

Return the strongest justified answer.

Thank You

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